

Assignment 2

due: 2026-09-23

Problem 1: If $X \subseteq G$ is a non-empty subset of G , then X is a subgroup if and only if $(*) \quad \forall a, b \in X, ab^{-1} \in X$.

Problem 2: Let $f : G \rightarrow H$ be a group homomorphism and let $S \subseteq G$ be any subset. Then

$$f(\langle S \rangle) \subseteq \langle f(S) \rangle.$$

Problem 3: Suppose that H, K are subgroups of G .

We say that H **normalizes** K if for each $g \in H$ we have $\text{Inn}_g K \subseteq K$ (in other words, $\forall x \in K, gxg^{-1} \in K$).

The **product** of H and K is the subset

$$HK := \{xy \mid x \in H, y \in K\}$$

.

Assume that H normalizes K . Prove that $\langle H, K \rangle = HK$. In particular, HK a subgroup of G .

Problem 4: Let H, K be subgroups of G and let $\varphi : H \times K \rightarrow HK$ be the natural mapping given by $\varphi(h, k) = hk$.

- For each $\alpha \in HK$, the set $\varphi^{-1}(\alpha)$ is in bijection with $H \cap K$.
- In particular, if $H \cap K = \{1\}$, then φ is bijective.

Problem 5: If G is a finite group and H, K subgroups of G , then

$$|HK| = |H| \cdot |K| / |H \cap K|.$$

Problem 6: Let $n \in \mathbb{N}, n > 0$ and consider the group $S = S(\mathbb{Z}/n\mathbb{Z})$ of permutations of the set $\mathbb{Z}/n\mathbb{Z}$.

For $x \in \mathbb{Z}/n\mathbb{Z}$, recall that the additive order $o(x)$ is a divisor of n .

- Describe the *cycle structure* of the element $\sigma \in S$ defined by the rule $\sigma(z) = z + x$. Show that the order of σ is $o(x)$.
- Suppose that $n = p$ is a prime number, and let $k \in \mathbb{Z}$ with $\gcd(k, p) = 1$. Thus the class $\bar{k}$ of k in $\mathbb{Z}/p\mathbb{Z}$ lies in the group $(\mathbb{Z}/p\mathbb{Z})^\times$ of *units*. The multiplicative order $o(\bar{k})$ of $\bar{k}$ is a divisor of $p - 1$. Describe the *cycle structure* of the element $\tau \in S$ defined by the rule $\tau(z) = \bar{k} \cdot z$. Show that the order of τ is $o(\bar{k})$.

Problem 7: Let G be a group and let Ω be a G -set. If $x, y \in \Omega$ and $x = gy$ for some $g \in G$, prove that the stabilizers $G_x = \text{Stab}_G(x)$ and $G_y = \text{Stab}_G(y)$ are *conjugate*. More precisely, show that

$$G_x = g \cdot G_y \cdot g^{-1}$$

Problem 8: Let G be a group. G acts on itself by conjugation: for $g, x \in G$, the action of g on x is given by $\text{Inn}_g x = gxg^{-1}$.

- Prove that the assignment $g \mapsto \text{Inn}_g$ determines a group homomorphism $G \rightarrow \text{Aut}(G)$ where $\text{Aut}(G)$ is the group of automorphisms of G .
- Let $Z = \{g \in G \mid \forall x \in G, gx = xg\}$ be the *center* of G . Prove that $Z = \ker \text{Inn}$.

For the action of G on itself by conjugation, the stabilizer $\text{Stab}_G(x)$ of $x \in G$ is usually written $C_G(x)$ and is called the *centralizer* of x in G . Note that

$$C_G(x) = \{y \in G \mid yxy^{-1} = x\} = \{y \in G \mid yx = xy\}$$

Problem 9: Let $I = I_n$ be a finite set with n elements, and let $S = S_n = S(I_n)$ be the group of permutations of I . Recall that $|S| = n!$.

- Prove that there are $(n-1)!$ n -cycles in S . **Hint:** If the elements of I are written $I = \{a_1, a_2, \dots, a_n\}$, then the n -cycles $(a_1, a_2, \dots, a_n)$ and $(a_2, a_3, \dots, a_n, a_1)$ are *equal*.
- Prove that if σ is an n -cycle in S , then $C_S(\sigma) = \langle \sigma \rangle$

Problem 10: Let $\sigma, \tau \in S = S(\mathbb{Z}/n\mathbb{Z})$ be defined by: $\sigma(i) = i + 1$, $\tau(i) = -i$. Notice or recall $\tau\sigma\tau = \sigma^{-1}$, so that

$$D := \langle \sigma, \tau \rangle = \langle \sigma \rangle \langle \tau \rangle$$

is a subgroup of S (using the fact that $\langle \tau \rangle$ normalizes $\langle \sigma \rangle$, together with Problem 3).

- Using Problem 5, show that

$$|D| = |\langle \sigma \rangle| \cdot |\langle \tau \rangle| / |\langle \sigma \rangle \cap \langle \tau \rangle|.$$

What are $|\langle \sigma \rangle|$ and $|\langle \tau \rangle|$?

- Show that if n is odd, $\langle \sigma \rangle \cap \langle \tau \rangle = \{1\}$, and conclude $|D| = 2n$ in this case.
- Now suppose $n = 2k$ is even. Identify the unique subgroup of order 2 in $\langle \sigma \rangle$, and show that τ is **not** that element – i.e. $\sigma^k \neq \tau$. (Hint: evaluate both sides at a well-chosen point of $\mathbb{Z}/n\mathbb{Z}$.)
- Conclude that $\langle \sigma \rangle \cap \langle \tau \rangle = \{1\}$ for all n , and hence $|D| = 2n$ always. (D is usually written D_{2n} , the dihedral group of order $2n$.)